

Daniel Carter

AI Application Engineer / Software Engineer

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Software Engineer with a BSc in Computer Science and practical experience developing backend applications and AI-powered software. Skilled in Python, REST API development, FastAPI, databases, Docker, Git, and AWS. Experienced in building and deploying LLM-powered applications and integrating external language model APIs into backend systems. Interested in scalable backend architecture and practical applications of generative AI.

Education

BSc Computer Science

University of Birmingham, UK | 2019–2022

Relevant coursework: Software Engineering, Database Systems, Web Development, Algorithms and Data Structures, Cloud Computing

Technical Skills

- **Programming:** Python
- **Backend Development:** REST APIs, FastAPI, Backend Development
- **AI / LLM:** LLM APIs, LLM Application Development, API Integration
- **Cloud:** AWS
- **Containers:** Docker
- **Databases:** Relational Databases, Database Integration, SQL
- **Development Tools:** Git, GitHub
- **Software Engineering:** API Design, Application Deployment, Testing, Version Control

Professional Experience

Software Engineer

Cloud Applications Ltd. | Birmingham, UK | 2022–2024

- Developed backend services and REST APIs using Python and FastAPI.
- Designed API endpoints for applications requiring database access and external service integrations.
- Worked with databases to store and retrieve application data.
- Used Git for source control and collaborative software development.
- Containerised applications using Docker to provide consistent development and deployment environments.
- Deployed backend applications and services to AWS. Worked on integrating third-party APIs into production applications.

Projects

LLM-Powered Application

- Built an application integrating an external LLM API to provide AI-powered functionality.
- Developed a Python backend to handle user requests and communicate with the language model API.
- Designed REST API endpoints for interacting with the application.
- Implemented error handling and response processing for external API requests.
- Containerised the application using Docker.
- Deployed the application to AWS.

REST API Development

- Designed and implemented RESTful APIs using Python and FastAPI.
- Created endpoints for application data retrieval, creation, and updates.
- Integrated APIs with a database for persistent data storage.
- Tested API functionality and handled common API errors and validation requirements.

- Used Git for version control throughout development.

AWS Application Deployment

- Deployed Python-based backend applications to AWS.
- Configured application environments and deployment workflows.
- Used Docker to package applications for deployment.
- Monitored deployed services and resolved application-level deployment issues.

Additional Experience

Software Engineering Intern

Tech Solutions Group | Birmingham, UK | 2021–2022

- Assisted with Python backend development and API implementation.
- Worked with databases and SQL queries as part of application development.
- Used Git-based workflows for managing source code.
- Supported testing and debugging of backend services.
- Assisted with deployment and configuration of development applications.

Certifications

- AWS Certified Cloud Practitioner Python Programming — Online Certification
- Docker Fundamentals — Online Certification

Interests

Artificial Intelligence, Generative AI, Backend Engineering, Cloud Computing, Software Architecture, Developer Tools